ANALYSIS OF SIMULATION OUTPUT

8.1 Estimation Methods:

Statistical estimation methods are commonly used to estimate parameters from sample observations on random variables. Usually, a random variable is drawn from an infinite population that has stationary probability distribution with a mean, μ and finite variance, σ^2. It means that population distribution is not affected by number of the samples being made, nor does it change with time. If the value of one sample is not affected by the value of other, the random variables are mutually independent.

The random variables which satisfy both of these conditions are said to be independently and ~~and~~ identically distribute (i.i.d.). The central limit theorem then could be applied for i.i.d. variables for simulation data. The CLT states that, "The sum of n i.i.d. variables drawn from a population that has a mean of μ and a variance of σ^2, is approximately distributed as a normal variable with a mean, $n\mu$ and a variance, $n\sigma^2$."

Any normal distribution can be transformed into a standard normal distribution, that has a mean 0 and a variance of 1. Let x_i $(i = 1, 2, \ldots, n)$ be the n i.i.d. random variables. Using CLT and applying transformation gives the following approximate normal variate:

$$z = \frac{\sum_{i=1}^{n} x_i - n\mu}{\sqrt{n}\,\sigma}$$

Divide both numerator & denominator by 'n'

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}, \quad \because \bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i = \frac{\text{Sum of } n \text{ i.i.d. RV}}{n}$$

The variable $\bar{x}$ is sample mean, which is sum of the random variables. So confidence interval about its computed value, needs to be established being a R.V.

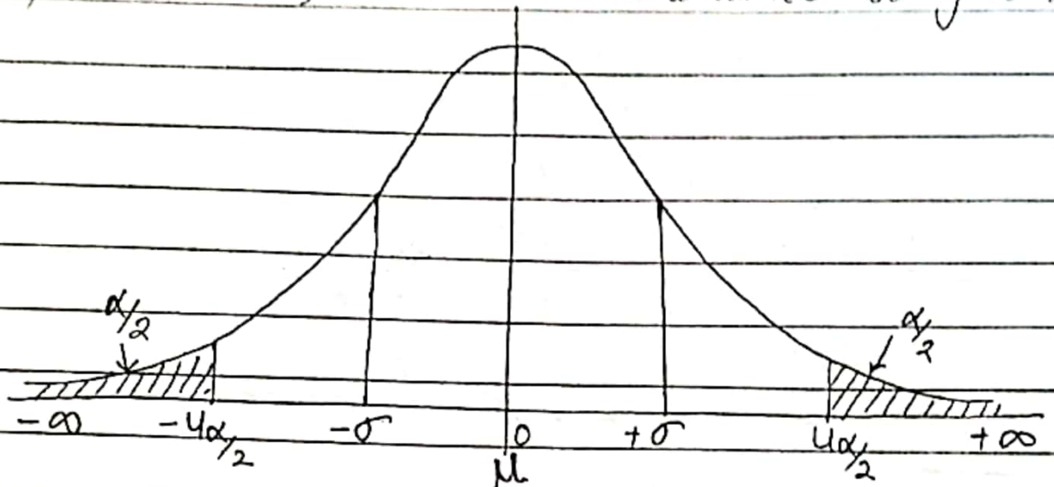

fig: prob. density function of standard normal variate

The above pdf of SNV shows the region in which z lies.

The integral from $-\infty$ to the value u is prob. that z is less than or equal to u, dentod by $\phi(u)$

Suppose $\phi(u) = 1 - \frac{\alpha}{2}$, α is < 1

then value of u is denoted by $u_{\alpha/2}$.

The prob that z is greater than $u_{\alpha/2}$ is then $\frac{\alpha}{2}$

The normal distribution is symmetric about its mean. so prob. that z is less than $-u_{\alpha/2}$ is also $\frac{\alpha}{2}$.

Consequently, the prob. that z lies betn $-u_{\alpha/2}$ & $u_{\alpha/2}$ is $1-\alpha$

i.e. $\text{Prob}\{-u_{\alpha/2} < z < u_{\alpha/2}\} = 1-\alpha$

In terms of sample mean,

Date ______
Page ______

$$\text{Prob}\left\{ \bar{x} + \frac{\sigma}{\sqrt{n}} u_{\alpha/2} \geq \mu \geq \bar{x} - \frac{\sigma}{\sqrt{n}} u_{\alpha/2} \right\} = 1-\alpha$$

- The constant $(1-\alpha)$ usually expressed as percent is confidence level and the interval $\left(\bar{x} \pm \frac{\sigma}{\sqrt{n}} u_{\alpha/2}\right)$ is confidence limit.

Typically CL is 90%, so $u_{\alpha/2} \geq 1.65$

$$\mu \Rightarrow \bar{x} \pm \frac{1.65\,\sigma}{\sqrt{n}}, \text{ prob.} = 90\%$$

The sample variance of estimate is

$$s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2$$

The normalized random variable based on σ^2 is replaced by normalized random variable based on s^2, which has Student-t distribution with $n-1$ dof.

The quantity $u_{\alpha/2}$, used in definition of CL is replaced by similar qty $t_{n-1,\alpha/2}$.

This Student-t distribution is strictly accurate only when popn from samples are drawn ~~randomly~~ is normally distributed.

Expressed in terms of estimated variance, s^2; the confidence interval for $\bar{x}$ is defined by

$$\boxed{\bar{x} + \frac{s}{\sqrt{n}} t_{n-1,\alpha/2}}$$

Hence, the above method shows the assumption of normality inherent in the use of central Limit Theorem (CLT) and hence, the method of establishing confidene intervals in terms of SNV, z and mean, μ.

8.2 Simulation Run Statistics

In simulation run statistics, it is assumed that the observations of samples are mutually independent and distribution from which they are drawn is also stationary.

Consider a single server system (M/M/1) in which arrivals occur with a Poisson distribution and service time has exponential distribution (M), with FIFO as the queuing discipline

The study objective is to measure mean waiting time defined as the time entities spend waiting to receive service and excluding service time itself.

This system is denoted by M/M/1, which indicates; first that inter-arrival time is distributed with Poisson's distribution (M); second service time is distributed exponentially; third that there is one server.

In a simulation run, simplest approach is to estimate mean waiting time by accumulating the waiting time of 'n' successive entities & dividing by 'n'. This measure, the sample mean, $\bar{x}(n)$ to emphasize the fact that its value depends upon number of observations taken.

If x_i $(i=1,2,\cdots n)$ are individual waiting times (including the value '0' for those entities which do not have to wait), then

$$\bar{x}(n) = \frac{1}{n}\sum_{i=1}^{n} x_i$$

The waiting time of each entities depends upon the waiting time of ~~its~~ their predecessors in a line, so the series of data are autocorrelated as one value affect other.

The sample mean of autocorrelated data can be shown to approximate a normal distribution as sample size increases. So the above mean $\bar{x}(n)$ of eqⁿ (1), remains a satisfactory estimate for the mean of autocorrelated data.

The variance is underestimated in M/M/1 system & the distribution may not be stationary.

Simulation run is started with system in some initial state, frequently idle state, in which no service is being given and no entities are waiting. The early arrivals than have a more than normal probability of ~~occurrence~~ obtaining service quickly, so sample mean which includes the early arrivals will be biased. As the length of the simulation run increases, the sample size increases and then effect of bias will die out.

For the sample size starting from a given initial condition, the sample mean distribution is stationary but if the distribution could be compared for different sample sizes, the distribution would be slightly different

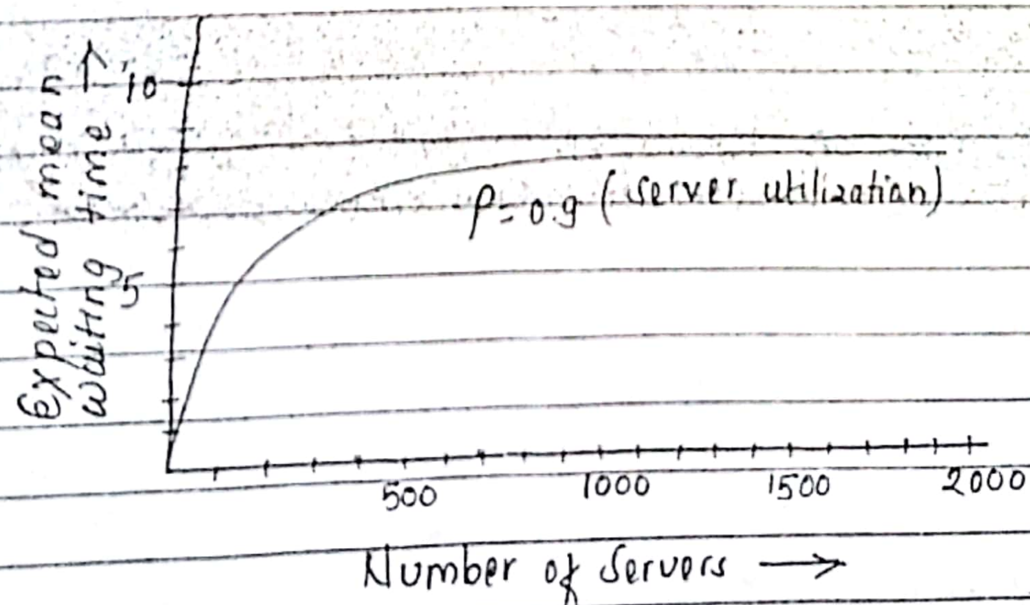

fig: Mean wait time in M/M/1 system for different sample sizes.

The above figure represent theoritical results, which shows how the expected value of sample mean depends upon the sample length.
For M/M/1 system, starting from an initial empty state, server utilization ie $\rho = \frac{\lambda}{\mu}\left(\frac{\text{arrival rate}}{\text{service rate}}\right) = 0.9$

The bias diminishes with increasing sample size as the utilization is also high. High utilization represents the performance in simulation of a run.

Replication of Runs:

The independent results could be obtained by repeating the simulation. Repeating the experiment with different random numbers for the same sample size n gives the set of independent determination of sample mean, $\bar{x}(n)$. This is replication of runs.

Even though distribution of sample mean depends upon degree of autocorrelation, these independent determinations of sample mean can be properly used to estimate variance of distribution. The precision of results of dynamic stochastic run be increased by repeating the experiments with the different R.N. strings.

Consider the experiment is repeated 'p' times with independent random number series. Let x_{ij} be the ith observation in jth run, and mean sample and variance for jth run be denoted by $\bar{x}_j(n)$ and $s_j^2(n)$, respectively.
then, for jth run, estimates are:

$$\bar{x}_j(n) = \frac{1}{n}\sum_{i=1}^{n} x_{ij}$$

$$s_j^2(n) = \frac{1}{n-1}\sum_{i=1}^{n}\left[x_{ij} - \bar{x}_j(n)\right]^2$$

— (I)

Combining the results for p independent measurements gives the following estimates for the mean waiting time, $\bar{x}$ and variance s^2, of the population

$$\bar{x} = \frac{1}{p}\sum_{j=1}^{p} \bar{x}_j(n)$$

$$s^2 = \frac{1}{p}\sum_{j=1}^{p} s_j^2(n)$$

— (II)

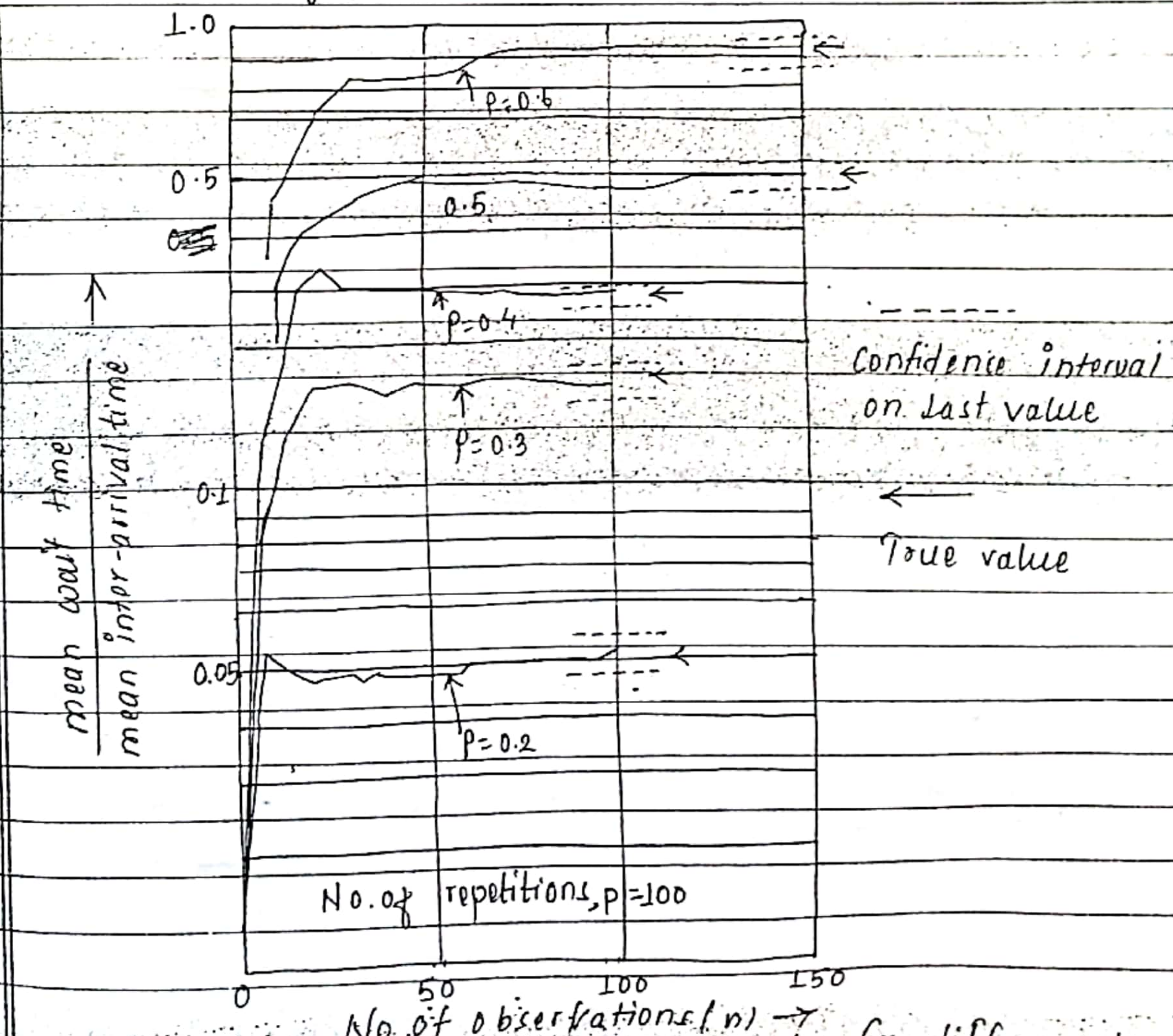

fig: Experimentally measured WT in M/M/1, for diff sample siz

The value of $\bar{x}$ is the estimate for mean waiting time and s^2 can be used to establish a confidence interval as in (II), with $p-1$ degree of freedom.

The above figure shows result of applying procedures to experimental results for M/M/1 system. Results are shown for server utilization 0·1, 0·2, 0·3, 0·4, 0·5 and 0·6.

In each case, the experiment been repeated from an initial idle state with different random numbers being used in each repetition.

The result shows the estimated mean waiting time as a function of sample size, n.

when $n=5$ for $\rho = 0.2, 0.3$ & 0.4

$n=10$ for $\rho = 0.5$ & 0.6

Each case is for 100 repetitions ($p=100$)

90% confidence interval for highest n

p repetitions of n ~~num~~ observations involve (total observations) $N = p.n$

The confidence intervals could be narrowed by increasing the number of replications & shortening length of each run.

But shortening of length of replicated runs, starting condition will increase. So keep the number of repetition as low as possible to approximate a normal distribution with the sample means.

VVI

Elimination of Initial Bias

Date
Page

Initial Bias:

When a simulation run is started in some initial conditions(state), frequently the idle state, in which no service is being given and no entities are waiting, then the early ~~a~~ arrivals have a more than a normal probability of obtaining service quickly. So the sample mean that includes early arrivals will be biased. Hence, the inclination of higher probability of obtaining the service due to no entities waiting initially or in the idle state in queue is called initial bias.

As the length of simulation run is extended and the sample size increases, the biasing effect will decrease.

Elimination:

Statistical biasing results from an unfair sampling of a population or from an estimation process which doesnot give accurate results on average. So in a simulation run, average waiting time is the sum of the waiting time of all samples, some customers donot have to wait being served in a queue, starting from initial or ~~st~~ idle state. So, such condition creates bias & error.

There are two approach of reducing point estimator bias by using artificial and unrealistic initial conditions in a steady state simulation:-

(i) Initialize the simulation in a state that is more representative of long run conditions.
eg. use a set of real data as initial conditions.

(ii) Divide the simulation in two phases i.e. warm up phase and steady state phase. Data collection doesn't start until the simulation passess warm up phase, i.e. ignore first part of simulation.

Scanned with CamScanner

The ideal situation is to know the steady state distribution for the system and select the initial condition from that distribution.
The repeated experiments on M/M/1 system, supplying an initial waiting line for each run, selected at the random from known steady state distribution of the waiting lines.
The common approach is to remove initial bias is to eliminate initial selection of run. The run is started from idle state and stopped after certain period of time. The run is restarted with statistics being gathered from point of restart. So its usual in simulation that statistics are gathered from the beginning & wipe out statistics gathered upto point of restart.
The disadvantage of eliminating first part of a simulation run is that the estimate of variance needed to establish a confidence limit must be based on less information.
So reduction in bias is obtained at the price of increasing the confidence interval size.

Q. Define initial bias. How it can be eliminated in simulation output analysis.
Q. How can you use replication of runs in analysis of simulation output? Explain with example.
Q. What is initial bias? What is approach for elimination of initial bias?
Q. How can you use estimation methods in an analysis of simulation output? Explain with example.